Statistical physics approach to global optimization of city traffic

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Abstract

Simple models are proposed to study the optimization of city traffic. Phase diagrams are discussed for these models.

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In recent decades, vehicular traffic flow has attracted a huge attention in a community of statistical physics. For freeway traffic, theoretical and numerical studies have been often investigated using cellular automata (CA) models [*e.g.*, Rule 184 and the Nagel-Schreckenberg (NaSch) model] and follow-the-leader models [*e.g.*, car-following model and optimal velocity (OV) model]. Meanwhile, the Biham-Middleton-Levine (BML) model and its variants have been used for studies of city traffic. BML model is a two-dimensional analogue of Rule 184 model and it shows phase transitions among free-flowing, jammed, and intermediate phases. In addition, NaSch model with a intersection is also studied towards optimal signalization of urban networks.

Our motivation of this study is to understand physics of global optimization of city traffic. While many previous studies was investigated for traffic control, as mentioned in the previous paragraph, only local controls were considered in those studies. They fixed routes of vehicles, and controlled signals to keep free-flowing phases for streets as far as possible. However, nowadays, intelligent transport systems are being developed. We believe better traffic optimization can be achieved if travel information of vehicles are aggregated to central controll system and the system gives appropriate route to each vehicle.

First of all, we consider a model of N parallel NaSch streets under open boundary condition. While probabilities of vehile extinction, β_i, are the same $\beta_i = \beta$, probabilities of vehicle injection α_i vary temporary according to states of streets. We define cluster of slow vehicles as sequential vehicles whose speed v is below the maximum $v_{\max}$ and measure the largest cluster size S_i for each street. Then we give α_i as

$$\alpha_i = \min\left\{ \frac{S_{\max} + a - S_i}{\sum_i S_{\max} + a - S_i} N\alpha, 1 \right\}, \qquad (1)$$

where $S_{\max} = \max_i S_i$, α is probability of vehicle injection per street, and a is an "inversed sensitivity" parameter. We measure the "order parameter" which is defined the number density of vehicles whose speed is equal to $v_{\max}$. Results are shown in Fig. 1.

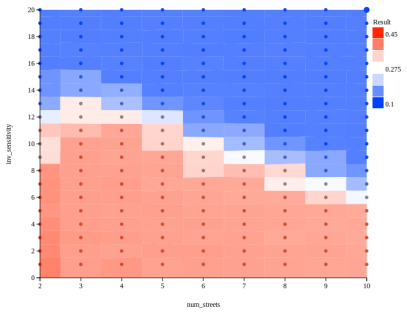


Figure1: Phase diagram of parallel NaSch model. Horizontal and vertical axes are inversed sensitivity a and the number of streets N, respectively. We set $\alpha = 0.35$, $\beta = 0.65$, $v_{\max} = 3$, and street length $L = 10000$. Color code shows the order parameter.

Furthermore, we consider a more realistic but still simple model. The system is a square grid graph under periodic boundary condition. Each vehicle is assumed that it can move with constant speed or stops, and has information about position of origin, departure time, and position of destination. If we consider space-time diagram for these vehicles, routes of vehicles can be regarded as a kind of directed polymers. Then the problem would become to search a ground state of the system whose Hamiltonian is $H = \sum_i R_i + J \sum_i c_{ij}$, where R_i is length of route i, c_{ij} is 1 if vehicles i and j are on the same street and 0 otherwise, and J is a parameter representing how easily street becomes jammed. We will discuss the results of Monte Carlo simulations of this model.